

## LUNGO, Sweden

WMO: 023820

Lat: 62.642N

Lon: 18.090E

Elev: 18

StdP: 101.11

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -17.5      | -14.8 | -20.2                         | 0.6 | -17.3 | -17.6 | 0.8 | -14.5 | 19.2                  | -5.5 | 17.3 | -3.5 | 5.0                   | 270  | 0.762 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.1                    | 23.2            | 18.2 | 21.5 | 17.3 | 20.1 | 16.4 | 19.6                | 22.0 | 18.3 | 20.6 | 17.1 | 19.2 | 5.0                  | 30   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.5                            | 13.4 | 21.3 | 17.3 | 12.4 | 19.7 | 16.2 | 11.5 | 18.4 | 56.1          | 22.1 | 51.8 | 20.7 | 48.3 | 19.2 | 23.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 16.1          | 14.0 | 12.2 | DB                         | -19.3 | 26.3               | 4.8 | 2.0                                                | -22.8 | 27.7       | -25.6 | 28.9       | -28.2 | 30.0       | -31.7 | 31.4 | (4) |
| (5)               |      |      | WB                         | -19.6 | 20.0               | 3.9 | 1.5                                                | -22.4 | 21.1       | -24.7 | 21.9       | -26.9 | 22.8       | -29.8 | 23.9 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 5.5   | -3.3  | -3.8  | -0.8  | 3.1   | 8.0   | 12.9  | 16.5  | 15.6  | 11.4  | 5.9   | 1.7   | -1.6 | (6)  |
| (7)  |                                                                                  | DBStd       | 7.92  | 4.90  | 5.54  | 4.18  | 2.74  | 3.00  | 2.65  | 2.51  | 2.46  | 2.68  | 3.00  | 3.70  | 4.81 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2180  | 412   | 386   | 334   | 207   | 77    | 5     | 0     | 0     | 17    | 132   | 249   | 361  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4697  | 671   | 619   | 592   | 457   | 321   | 163   | 70    | 91    | 209   | 387   | 499   | 619  | (9)  |
| (10) |                                                                                  | CDD10.0     | 544   | 0     | 0     | 0     | 0     | 16    | 92    | 201   | 175   | 58    | 3     | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 20    | 0     | 0     | 0     | 0     | 0     | 1     | 13    | 7     | 0     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 45    | 0     | 0     | 0     | 0     | 2     | 1     | 32    | 10    | 0     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 4     | 0     | 0     | 0     | 0     | 1     | 0     | 3     | 1     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 5.5   | 6.2   | 6.0   | 5.6   | 5.1   | 5.0   | 5.0   | 4.8   | 5.0   | 5.2   | 5.8   | 6.1   | 6.2  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 717   | 53    | 40    | 46    | 46    | 45    | 48    | 62    | 75    | 79    | 77    | 88    | 63   | (15) |
| (16) |                                                                                  | PrecMax     | 1072  | 131   | 117   | 127   | 142   | 134   | 146   | 165   | 199   | 211   | 226   | 260   | 278  | (16) |
| (17) |                                                                                  | PrecMin     | 445   | 1     | 6     | 3     | 3     | 12    | 4     | 5     | 12    | 6     | 7     | 17    | 4    | (17) |
| (18) |                                                                                  | PrecStd     | 131   | 32    | 26    | 28    | 34    | 25    | 29    | 39    | 47    | 44    | 48    | 57    | 45   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 5.8   | 7.3   | 10.9  | 13.6  | 19.5  | 22.9  | 26.1  | 24.6  | 19.6  | 13.7  | 11.1  | 7.5  | (19) |
| (20) |                                                                                  |             | MCWB  | 3.9   | 4.1   | 6.0   | 7.9   | 12.4  | 14.9  | 19.1  | 19.5  | 13.9  | 10.6  | 8.1   | 5.2  | (20) |
| (21) |                                                                                  | 2%          | DB    | 4.2   | 5.9   | 8.1   | 10.7  | 16.2  | 20.4  | 24.0  | 22.3  | 17.5  | 12.1  | 8.2   | 5.7  | (21) |
| (22) |                                                                                  |             | MCWB  | 2.7   | 3.3   | 4.5   | 6.4   | 11.1  | 14.2  | 19.0  | 18.4  | 14.0  | 10.4  | 7.0   | 4.1  | (22) |
| (23) |                                                                                  | 5%          | DB    | 3.3   | 4.1   | 6.1   | 9.0   | 14.3  | 18.6  | 22.2  | 20.8  | 16.2  | 11.1  | 7.1   | 4.7  | (23) |
| (24) |                                                                                  |             | MCWB  | 2.2   | 1.9   | 3.1   | 5.3   | 9.8   | 13.9  | 18.0  | 17.5  | 13.4  | 9.8   | 6.3   | 3.6  | (24) |
| (25) |                                                                                  | 10%         | DB    | 2.5   | 2.5   | 4.5   | 7.3   | 12.6  | 17.2  | 20.6  | 19.6  | 15.1  | 10.1  | 6.3   | 3.9  | (25) |
| (26) |                                                                                  |             | MCWB  | 1.6   | 1.2   | 2.1   | 4.5   | 8.9   | 13.0  | 17.2  | 16.5  | 13.0  | 8.9   | 5.5   | 2.9  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 4.2   | 4.6   | 6.5   | 8.3   | 13.7  | 16.7  | 21.4  | 20.8  | 16.0  | 12.0  | 8.4   | 5.6  | (27) |
| (28) |                                                                                  |             | MCDWB | 5.1   | 6.8   | 9.9   | 12.3  | 17.9  | 19.8  | 24.5  | 22.8  | 17.6  | 12.7  | 10.0  | 6.8  | (28) |
| (29) |                                                                                  | 2%          | WB    | 3.2   | 3.5   | 4.7   | 6.9   | 11.7  | 15.6  | 20.0  | 19.4  | 15.1  | 11.0  | 7.3   | 4.6  | (29) |
| (30) |                                                                                  |             | MCDWB | 3.9   | 5.5   | 7.6   | 10.0  | 15.3  | 18.6  | 22.6  | 21.5  | 16.3  | 11.6  | 8.0   | 5.3  | (30) |
| (31) |                                                                                  | 5%          | WB    | 2.4   | 2.3   | 3.3   | 5.8   | 10.5  | 14.6  | 18.8  | 18.3  | 14.3  | 10.1  | 6.5   | 3.8  | (31) |
| (32) |                                                                                  |             | MCDWB | 3.1   | 3.5   | 5.6   | 8.4   | 13.4  | 17.6  | 21.5  | 20.2  | 15.5  | 10.9  | 7.0   | 4.5  | (32) |
| (33) |                                                                                  | 10%         | WB    | 1.6   | 1.5   | 2.4   | 4.8   | 9.3   | 13.7  | 17.6  | 17.2  | 13.5  | 9.0   | 5.7   | 3.0  | (33) |
| (34) |                                                                                  |             | MCDWB | 2.3   | 2.4   | 4.1   | 6.9   | 11.9  | 16.5  | 20.0  | 18.9  | 14.7  | 10.0  | 6.1   | 3.7  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 4.6   | 5.2   | 5.9   | 5.8   | 6.3   | 6.6   | 6.1   | 6.0   | 5.4   | 4.5   | 3.6   | 4.3  | (35) |
| (36) |                                                                                  |             | MCDBR | 4.2   | 6.0   | 7.5   | 9.0   | 9.1   | 8.6   | 8.5   | 7.9   | 6.7   | 5.3   | 3.9   | 4.5  | (36) |
| (37) |                                                                                  |             | MCWBR | 3.7   | 4.5   | 4.9   | 5.5   | 5.1   | 4.5   | 4.8   | 4.7   | 4.2   | 3.9   | 3.4   | 3.9  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWR | 3.9   | 5.4   | 6.8   | 8.5   | 8.4   | 7.4   | 7.3   | 6.8   | 5.5   | 4.4   | 3.6   | 4.2  | (38) |
| (39) |                                                                                  |             | MCWBR | 3.6   | 4.3   | 4.6   | 5.4   | 5.2   | 4.4   | 4.4   | 4.4   | 4.0   | 3.7   | 3.3   | 3.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.240 | 0.297 | 0.318 | 0.355 | 0.355 | 0.350 | 0.369 | 0.361 | 0.329 | 0.314 | 0.257 | 0.345 | (40) |      |
| (41) |                                                                                  | taud        | 2.109 | 2.244 | 2.340 | 2.370 | 2.443 | 2.494 | 2.463 | 2.479 | 2.551 | 2.435 | 2.168 | 2.598 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 420   | 650   | 789   | 830   | 863   | 874   | 844   | 820   | 785   | 638   | 390   | 252   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 32    | 61    | 81    | 97    | 98    | 95    | 96    | 87    | 67    | 51    | 30    | 19    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.17  | 0.75  | 2.23  | 3.80  | 5.09  | 5.84  | 5.35  | 4.06  | 2.47  | 1.03  | 0.27  | 0.09  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.03  | 0.14  | 0.26  | 0.35  | 0.50  | 0.36  | 0.52  | 0.31  | 0.27  | 0.18  | 0.06  | 0.01  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     | (n)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (36 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page